

Instructions: There are totally 1 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Let $f(x) = \frac{1}{4}x^4 - 2x^3 + \frac{9}{2}x^2 + 5$.

(a) Find all critical points of f .

$$\begin{aligned} f'(x) &= x^3 - 6x^2 + 9x \quad \rightarrow \text{differentiable everywhere.} \\ &= x(x^2 - 6x + 9) \\ &= x(x-3)^2 \end{aligned}$$

Thus, $f'(x) = 0$ at $x=0$ and 3 . Both of them are critical points of f .

(b) Find all relative extrema of f using the second derivative test.

$$\begin{aligned} f''(x) &= 3x^2 - 12x + 9 \\ &= 3(x^2 - 4x + 3) \end{aligned}$$

$$f''(0) = 3 \cdot (0 - 0 + 3) = 9 > 0, \text{ so } f \text{ has a relative minimum } f(0) = 5 \text{ at } x=0$$

$$f''(3) = 3(9 - 12 + 3) = 0, \text{ so it is inconclusive at } x=3.$$

(Remark: you can find f doesn't have a relative extremum at $x=3$)

(c) Find the absolute maximum and minimum values of f on the interval $[-1, 4]$. by the first derivative test)

Evaluate f at all critical points and the endpoints.

$$f(0) = 5$$

$$f(3) = \frac{1}{4} \cdot 3^4 - 2 \cdot 3^3 + \frac{9}{2} \cdot 3^2 + 5 = 11\frac{3}{4}$$

$$f(-1) = \frac{1}{4}(-1)^4 - 2 \cdot (-1)^3 + \frac{9}{2} \cdot (-1)^2 + 5 = 11\frac{3}{4}$$

$$f(4) = \frac{1}{4} \cdot 4^4 - 2 \cdot 4^3 + \frac{9}{2} \cdot 4^2 + 5 = 13$$

Thus f has a absolute minimum 5 at $x=0$ and a absolute maximum 13 at $x=4$ on the interval $[-1, 4]$